

Automatic Chromosome Detection and Classification Using YOLOv8 and Transfer Learning

Ch. Rambhadra Kumar

Roll No : 202311024

B.Tech Computer Science and Engineering

Indian Institute of Information Technology Vadodara

Summer Internship Report 2026

202311024@diu.iiitvadodara.ac.in

Under the Guidance of Dr. Venkata Phani Krishna B

Abstract—Cytogenetic diagnosis relies heavily on chromosome analysis to detect genetic abnormalities and chromosomal disorders. But Manual analysis, however, is slow, and it mainly depends on the availability of experienced cytogeneticists. This report presents a Two stage deep learning pipeline, developed during the internship, for automating chromosome detection and classification from full human karyotype images (consists of mixed 24 classes). In which the first stage uses a YOLOv8 object detector, trained on full karyotypic images along with their XML annotations, to locate and box individual chromosomes in a photo of mixed chromosomes. The detected regions are then cropped automatically and passed as input to the deep transfer learning classifiers like ResNet50, EfficientNet, InceptionV3, and CIRNet which classify and sort them into the twenty-four standard chromosome classes (1 to 22, X, and Y). The same XML annotations which are used to train the detector also used in both the localization and classification stages. In the experiments conducted, the YOLO detector reached roughly about 80% accuracy for detection, and the classification models gave around 90% accuracy on the cropped single chromosome images. Although the project finished room for improvement remains, but the resulting pipeline cuts down a substantial manual effort and offers a workable path gives a more automated chromosome analysis.

Index Terms—Chromosome Detection, Chromosome Classification, YOLO, Deep Learning, Transfer Learning, ResNet50, EfficientNet, CIRNet, Medical Image Analysis, Cytogenetics.

I. INTRODUCTION

Every cell in a human body carries trillions of chromosomes, and the way they are arranged and structured tells us a great deal about a person's genetic health. Cytogenetic experts routinely examine metaphase chromosome images and arrange them into a standard karyotype to catch and identify conditions such as Down syndrome, Edwards syndrome, Patau syndrome, and a range of structural chromosome disorders. This process, even though dependable in trained hands, but is a boring and lasting for very long, time consuming, and also not immune to disagreement between different observers looking at the same image.

Over the last few decades, deep learning has made a significant progress in medical image analysis. Now Convo-

lutional Neural Networks (CNNs) are quite good at learning discriminative visual features straight from raw pixels, without anyone having to hand craft those features manually. On a related place, object detectors like YOLO (You Only Look Once) can locate and detect several objects in a single image while still running fast enough for practical use.

A recurring pattern in the published literature on chromosome classification is taken for the assumption that chromosomes arrive already segmented something, has cut each one of the images before the classifier ever sees them. That assumption breaks down in a real clinical setting, where the raw input is a full metaphase image (24 classes) and nothing has been separated first. This gap motivated the two stage design that we adopted here, where detection and classification were handled as two distinct and sequential steps rather than folded into a single model classification model.

The internship project began with full complete chromosome images, and run through a YOLO based detector to locate and find each of the chromosomes in the frame. Each detected chromosome was then cropped out and handed to a transfer learning classifiers such as ResNet50, EfficientNet, CIRNet, and an Inception based model were all trained to identify which class the chromosomes belongs to. Performance was measured by comparing the predicted labels against the XML annotations bundled with the dataset.

II. OBJECTIVES

The internship was guided by the following objectives:

- Build an automated pipeline that can detect chromosomes directly from complete human karyotype images.
- Train a YOLO-based detector using a dataset annotated in XML format.
- Automatically extract the detected chromosome regions for use in the next stage.
- Train and compare several transfer learning models – ResNet50, EfficientNet, CIRNet, and an Inception-based architecture – for classifying chromosomes into 24 classes.

- Check predicted labels against the XML ground truth to get a quantitative sense of performance.
- Understand, overall, how well detection and classification work together as a single automated system.

III. TASKS ASSIGNED DURING INTERNSHIP

The following tasks were assigned over the course of the internship:

- Study existing chromosome datasets and get familiar with chromosome morphology.
- Prepare the datasets so they could be used for both detection and classification.
- Convert the XML annotations into YOLO's annotation format.
- Train a YOLOv8 model to detect chromosomes.
- Train deep transfer learning models to classify the detected chromosomes.
- Wire the detection and classification stages together into one working pipeline.
- Evaluate the whole system against the XML ground truth.

IV. DATASET DESCRIPTION

Two datasets were used at different points in this project. For training the classifier, a set of individual chromosome images already sorted into the twenty-four chromosome classes (1–22, X, and Y) was used. These gave the classification models a clean, focused way to learn what each chromosome type looks like.

For training the detector, full metaphase chromosome images paired with XML annotation were used. Each of these files records the chromosome's class label along with their bounding box coordinates.

At testing time, a complete metaphase image was given to the classifier with no prior segmentation. YOLO detector finds all the chromosomes in the image, then crops each chromosome automatically, and then passes the cropped chromosomes to the trained classifier for identification.

V. PROPOSED FRAMEWORK

There are two consecutive stages:

- 1) Detection of chromosomes Using YOLO from a complete metaphase image.
- 2) Classification of the cropped chromosomes Using deep learning models.

Figure 1 shows the full processing pipeline.

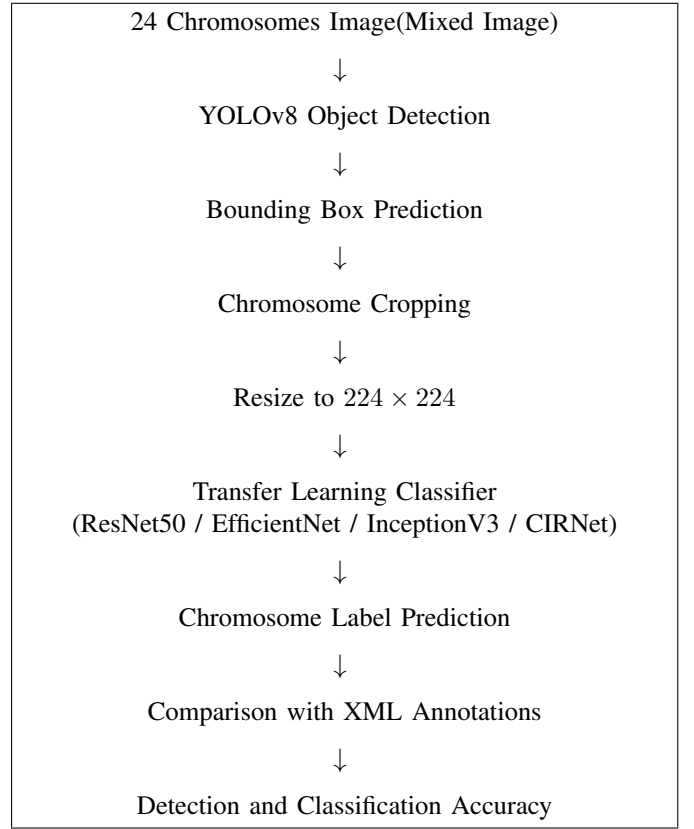


Fig. 1: Overall workflow of the proposed chromosome detection and classification framework.

VI. METHODOLOGY

The pipeline runs as a twomstage process, starting from giving complete, unsegmented 24 chromosome image. Rather than assuming someone has already cropped out class individual chromosomes like most classification only methods perform ,the system takes the raw karyotype image as input then a YOLOv8 detector finds and localizes each and every chromosome first, and then classification happens afterward, using deep transfer learning models. Below are the stages involved.

A. Stage I: Chromosome Detection Using YOLOv8

B. Stage II: Chromosome Classification Using

- ResNet50
- EfficientNet
- InceptionV3
- CIRNet

$\{1, 2, 3, \dots, 22, X, Y\}$

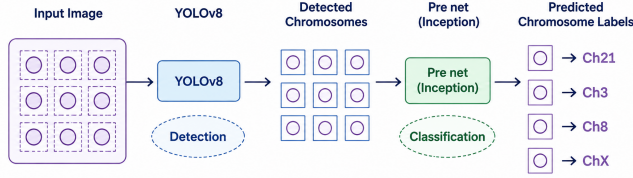


Fig. 2: Two stage chromosome detection and classification workflow.

VII. WORK CARRIED OUT

Work on this internship began with preprocessing the chromosome datasets. The XML annotation files were converted into YOLO's annotation format, and the data was split into training, validation, and testing subsets.

A YOLOv8 detector was trained to find and locate the chromosomes in complete metaphase images. Once trained, the detector predicts chromosome positions, and each one gets cropped out automatically with respect to their positions.

The cropped images tested through several transfer learning models – ResNet50, EfficientNet, InceptionV3, and CIRNet – with their predictions compared using the XML annotations for measuring classification performance.

The complete detection and classification pipeline were tested last, on full metaphase images that the models had not encountered before, to see how it would result in practice.

Tools and libraries used throughout the project are:

- TensorFlow
- Keras
- OpenCV
- NumPy
- Matplotlib
- Ultralytics YOLO
- XML ElementTree

Training data for the classifier consists of individual chromosome images of the twenty four classes, while testing was done on complete, unsegmented chromosome images. The XML files supplied with the dataset contains the reference labels required for the evaluation.

After splitting the dataset into training, validation, and test sets, the YOLO detector was trained on the converted YOLO annotations. During inference, the detector locates chromosomes of full metaphase image, and the resulting crops are passed on to the pretrained classification models for classification.

VIII. INDIVIDUAL CONTRIBUTIONS

Individual contribution completed over the course of this internship include:

- Preprocessing and organizing the dataset.
- Parsing the XML annotations and converting them into YOLO format.
- Training the YOLOv8 chromosome detector.
- Building the automatic chromosomes cropping pipeline.

- Training and evaluating ResNet50, EfficientNet, InceptionV3, and CIRNet.
- Implementing dl models with XML based evaluation process.
- Analyzing and reviewing the experimental results.
- Making the technical documentation and this report.

IX. RESULTS AND DISCUSSION

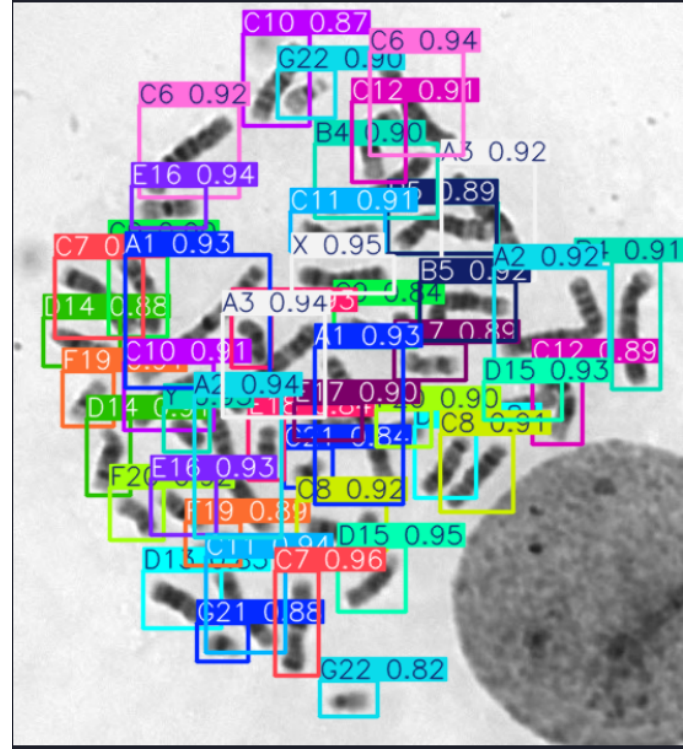


Fig. 3: YOLOv8 detection results on a sample metaphase chromosome image. and Colored bounding boxes indicating detected chromosomes.

Table I gives the final observed results.

TABLE I: Experimental Results

Model	Accuracy (%)
YOLOv8 Detection	80
ResNet50	91
EfficientNet	92
InceptionV3	89
CIRNet	92

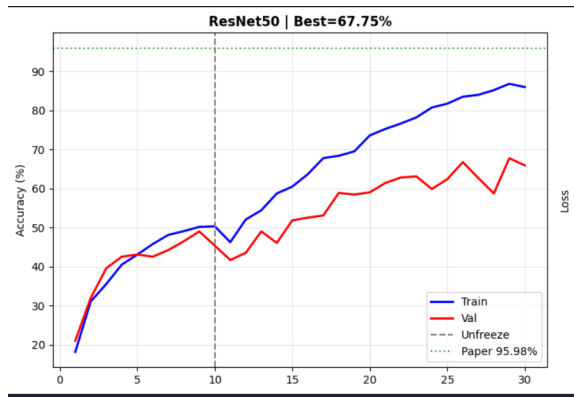


Fig. 4: Training and validation accuracy of ResNet50.

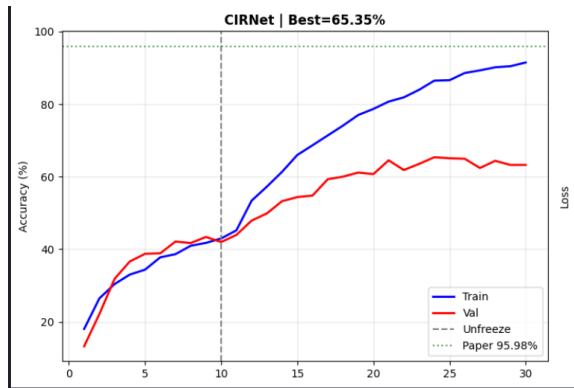


Fig. 5: Training and validation accuracy of CIRNet.

A. Per-Class Performance

Per-class results:								
Chr	True	Predicted	Normal	Abnormal	Avg Conf%	Precision	Recall	F1 Status
Chr 1	30	25	18	7	76.400002	0.840	0.700	0.764 ▲
Chr 2	30	21	9	12	63.000000	0.714	0.500	0.588 ▲
Chr 3	30	26	20	6	78.500000	0.692	0.600	0.643 ▲
Chr 4	30	41	30	11	79.099998	0.561	0.767	0.648 ▲
Chr 5	30	36	19	17	68.699997	0.528	0.633	0.576 ▲
Chr 6	30	49	21	28	63.200001	0.388	0.633	0.481 ✖
Chr 7	30	38	18	20	64.400002	0.579	0.733	0.647 ▲
Chr 8	30	38	11	27	57.200001	0.447	0.567	0.500 ▲
Chr 9	30	20	13	7	69.099998	0.650	0.433	0.520 ▲
Chr 10	30	23	9	14	59.099998	0.609	0.467	0.528 ▲
Chr 11	30	26	16	10	71.599998	0.731	0.633	0.679 ▲
...								
Chr 21	30	27	24	3	86.300003	0.963	0.867	0.912 ✓
Chr 22	30	33	27	6	79.599998	0.697	0.767	0.730 ▲
Chr X	29	34	11	23	57.799999	0.500	0.586	0.540 ▲
Chr Y	23	23	23	0	98.199997	1.000	1.000	1.000 ✓

Fig. 6: Per-class classification performance.

X. INTERNSHIP EXPERIENCE

This internship provided hands on experience to work on computer vision, deep learning, and medical image analysis that reading papers alone couldn't have offered. Working through chromosome detection and classification gave a much stronger intuition on object detection, transfer learning, dataset preprocessing, model learning and model evaluation.

Practical skills about Python, TensorFlow, PyTorch, OpenCV, deep learning models and YOLOv8 were sharpened along the way, along with real, hands on experience with

messy medical image dataset and XML annotations. Regular talks with the mentor proved genuinely useful, encouraging deeper thinking through problems and offered a clear picture of how deep learning workflows are trained and used together in practice.

XI. CONCLUSION

During this internship, a two stage deep learning framework was developed for detecting and classifying chromosomes from complete metaphase chromosome images(24 chromosomes). A YOLOv8 detector handles localization(Detecting chromosomes), while ResNet50, EfficientNet, InceptionV3, and CIRNet models were each used for the classification step.

The results indicate that pairing object detection with deep transfer learning is an effective way to automate chromosome classification. Using the existing XML annotations it became a straightforward way to evaluate both stages reliably. Room for improvement clearly remains, particularly on the chromosome detection side, but this framework became a solid starting point for future work on computer aided cytogenetic diagnosis.

XII. FUTURE PROSPECTS

Looking ahead, this framework can be used to flag chromosomal abnormalities like if a chromosome is broken (abnormal) or healthy. Accuracy in detecting chromosomes in particular can be pushed higher with larger object detection models and a broader range of training dataset.

Apart from the technical outcomes, this internship gave a solid understanding of foundations in artificial intelligence, computer vision, and medical image analysis, and it showed the directions worth pursuing further in future research more broadly.

REFERENCES

- [1] J. Redmon, S. Divvala, R. Girshick and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," *Proceedings of CVPR*, 2016.
- [2] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," *CVPR*, 2016.
- [3] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *ICML*, 2019.
- [4] C. Szegedy et al., "Rethinking the Inception Architecture for Computer Vision," *CVPR*, 2016.
- [5] M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," 2016.

T.B.V. Phaikurme